Feiyang Huang

PhD Candidate, Computational Biology & Medicine

New York, NY

English, Mandarin

Education

PhD **Weill Cornell Graduate School of Medical Sciences**, Tri-Institutional PhD Program in Computational Biology & Medicine — New York, NY — Aug 2023 – present
- Advisor: Wesley Tansey, Memorial Sloan Kettering Cancer Center

BS **Johns Hopkins University**, Computer Science and Biomedical Engineering — Baltimore, MD — Sept 2020 – May 2023
- GPA 3.96

Research Experience

Memorial Sloan Kettering Cancer Center, PhD Rotations — Aug 2023 – May 2024
- **Tansey Lab**: Bayesian active learning for nominating combination therapies from ex vivo drug screens.
- **Chodera Lab**: Equivariant graph neural networks for small-molecule property prediction (mtenn).
- **CMO Innovation Lab**: Computational pipeline for bisulfite- and enzyme-based DNA methylation sequencing of tumors.
- **Morris Lab**: BERT representation learning from tumor mutation profiles.

Johns Hopkins University, Fan Lab, Undergraduate Researcher — Jan 2021 – May 2023
- Reference-free deconvolution of spot-based spatial transcriptomics using LDA topic modeling.
- Quantification of subcellular mRNA polarization using a torque framework.

Johns Hopkins University, Institute for NanoBioTechnology, Research Intern — May 2021 – Aug 2021
Microfluidic chip platform for rapid (<10 min) point-of-care PCR diagnostics.

National University of Singapore, Yu Lab, Research Intern — Nov 2019 – Dec 2020
Lesion invasion depth prediction from endoscopic images using a CNN.

Honors and Awards

NIH DEBUT Challenge, Healthcare Technologies for Low-Resource Settings Prize — 2023

VentureWell e-Fest Prize — 2023

Preprints

Population-Scale Precision Safety in Oncology Reveals Clinical and Genetic Determinants of Systemic Therapy Toxicity — Sept 2026
Ziad Bakouny*, X Alex Guo*, **Feiyang Huang***, Saksham Mohan*, Rohan Walser*, et al., Wesley Tansey, Jian Carrot-Zhang, Ed Reznik
www.medrxiv.org/content/10.64898/2026.09.16.26363259 (medRxiv; *equal contribution)

A Pan-Cancer Ex Vivo Drug Screen Atlas for Functional Precision Oncology — Feb 2026
Karl Pichotta, Jessica B White, Jeffrey F Quinn, Anneliese Markus, Christopher Tosh, Antoine De Mathelin, Erin Coyne, **Feiyang Huang**, Wesley Tansey
pmc.ncbi.nlm.nih.gov/articles/PMC12934811 (bioRxiv)

Peer-Reviewed Publications

Reference-free cell type deconvolution of multi-cellular pixel-resolution spatially resolved transcriptomics data — Apr 2022
Brendan F Miller, **Feiyang Huang**, Lyla Atta, Arpan Sahoo, Jean Fan
10.1038/s41467-022-30033-z (Nature Communications 13, 2339)

Artificial intelligence-enhanced white-light colonoscopy with attention guidance predicts colorectal cancer invasion depth — Sept 2021
Xiaobei Luo, Jiahao Wang, Zelong Han, Yang Yu, Zhenyu Chen, **Feiyang Huang**, et al., Hanry Yu
10.1016/j.gie.2021.03.936 (Gastrointestinal Endoscopy 94(3), 627–638)

Roles and regulation of long noncoding RNAs in hepatocellular carcinoma — Oct 2019
Lee Jin Lim, Samuel YS Wong, **Feiyang Huang**, et al., Caroline G Lee
10.1158/0008-5472.CAN-19-0255 (Cancer Research 79(20), 5131–5139)

Conference Abstracts

Pan-cancer discovery of clinical and genomic determinants of immune-related adverse events 2026
Ziad Bakouny, X Alex Guo, Rohan Walser, **Feiyang Huang**, et al.
10.1200/JCO.2026.44.16_suppl.11132 (Journal of Clinical Oncology 44(16_suppl), 11132; ASCO 2026)

Development of a computational model of SBS scores in critically ill children 2024
S Wu, B Pejic, S Roy, **Feiyang Huang**, et al.
10.1097/01.pcc.0001085444.27374.df (Pediatric Critical Care Medicine 25(11S), e87)

Presentations

OncoState: Learning patient states from clinical trajectories in cancer Poster. 45th Vincent du Vigneaud Research Symposium, Weill Cornell Medicine, New York, NY. 2026

Towards point-of-care internal hemorrhage detection through a syndecan-1 lateral flow assay Oral presentation. Military Health System Research Symposium, Kissimmee, FL. 2022

Patents

Granted: US 12,105,099 B2; Methods and devices for quantitatively estimating syndecan-1

Pending: US 19/240,058; Methods and devices for quantitatively estimating thrombomodulin

Teaching

Course Assistant, Mathematical Foundations for Computer Science Johns Hopkins University Fall 2022

Leadership and Service

Singapore Students' Association, Johns Hopkins University, President 2021 – 2023

Theta Tau Professional Engineering Fraternity, Johns Hopkins University, Member 2020 – 2022

Work and Entrepreneurship Experience

DioTeX Diagnostics, Co-Founder Nov 2020 – Aug 2023
- Developed a lateral flow test for internal hemorrhage.
- Raised over $60,000 in nondilutive funding.

BME Design Team, Team Leader Aug 2022 – May 2023
- Developed a continuous monitoring solution using accelerometry and vitals for sedative dosage adjustment in the pediatric ICU.
- Led a team of 7 undergraduates and managed the project.
- Launched an IRB study to generate a dataset of regular interval, nurse annotated clinical sedation score.
- Obtained $100,000 in competitive internal grant funding.
- Interviewed 20 clinicians for needs discovery.

Hologic, Software Engineering Intern May 2022 – Aug 2022
- Built an internal work history search tool and automated release note generation.

Other Experience

Singapore Armed Forces, 3rd Sergeant Jan 2018 – Oct 2019
- Led a detachment of 3 soldiers and co-supervised daily training for 15 soldiers.